

Review

Transforming higher education with robotic process automation: enhancing efficiency, innovation, and student-centered learning

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Robotic Process Automation (RPA) is transforming the education sector by automating repetitive, rule-based tasks previously carried out by humans. This paper investigates the applications, benefits, challenges, and implications of RPA in education, focusing on its impact on administrative processes, student support services, and teaching methodologies. By analyzing 54 studies conducted between 2020 and 2024, this research explores how educational institutions have adopted RPA to enhance operational efficiency. Case studies reveal common success factors, including significant time savings, error reduction, and improved satisfaction for both students and staff. RPA has enabled institutions to streamline administrative workflows, optimize resource management, and offer personalized learning experiences. However, challenges such as implementation costs, data security, and the need for effective change management are also addressed. Additionally, this paper examines ethical concerns, particularly those surrounding job displacement and privacy. The findings underscore the transformative potential of RPA while highlighting the complexities of its integration into educational settings. The methodology followed PRISMA guidelines, with a systematic search of academic databases including IEEE Xplore, Scopus, Web of Science, and Google Scholar. Inclusion criteria focused on studies published between 2020 and 2024 that addressed RPA applications in education. Data analysis was conducted using thematic synthesis to identify key trends across the selected studies.

Keywords Robotic process automation (RPA) · Education sector · Intelligent automation · Administrative efficiency · Change management

1 Introduction

Education is undergoing a significant transformation in the digital age. With advancements in technology, traditional methods of teaching and learning are evolving to adapt to the needs and demands of the twenty-first century. One such technological innovation that has the potential to revolutionize education is robotic process automation (RPA). RPA, often associated with streamlining business processes in various industries, holds immense promise in the realm of education, offering opportunities to enhance administrative efficiency, improve teaching methodologies, and personalize learning experiences for students [1–3]. This paper provides a comprehensive overview of RPA in higher education by exploring its adoption across various educational institutions and processes. It addresses key research questions on the success

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factors and challenges of RPA adoption in higher education, its impact on administrative and teaching efficiency, the enhancement of student-centered learning experiences, and the ethical considerations and challenges institutions face in implementing RPA.

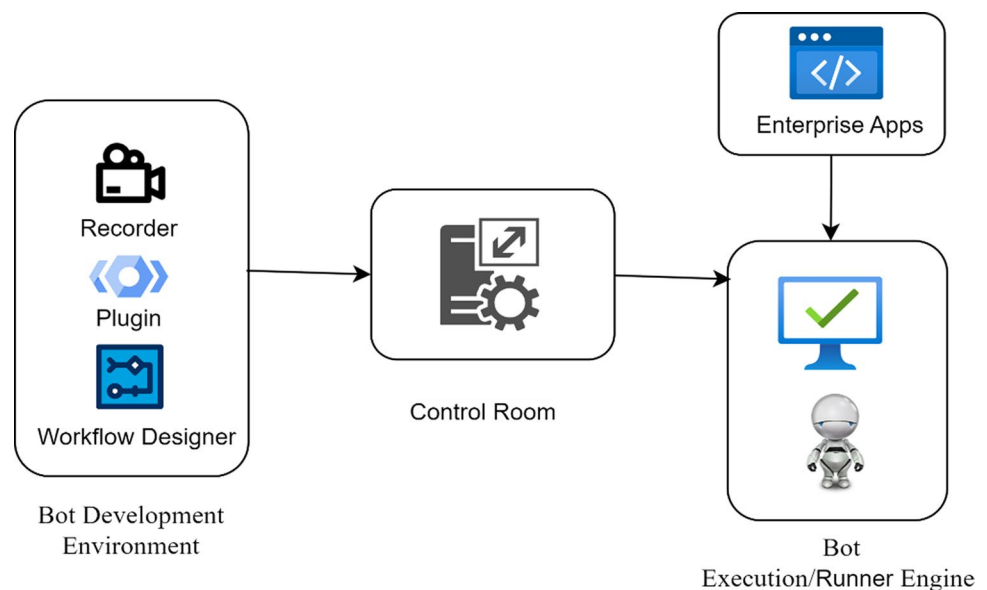
RPA is the automation of repetitive, rule-based processes that are traditionally handled by humans using software robots or “bots” [4]. These bots are programmed to mimic human actions within digital systems, interacting with software applications to execute tasks such as data entry, data manipulation, and form processing. RPA technology is characterized by its ability to automate mundane tasks with speed, accuracy, and scalability, thereby freeing up human resources to focus on more value-added activities. While RPA focuses primarily on rule-based tasks and structured data, it can be integrated with other intelligent automation technologies, such as artificial intelligence (AI), machine learning (ML), and natural language processing (NLP), to enable more advanced automation capabilities and handle a broader range of tasks [5, 6]. The evolution from traditional automation to intelligent automation represents a significant shift in how tasks are performed and managed in various industries [7, 8]. Traditional automation involves using machines, systems, or software to execute repetitive tasks with minimal human intervention. These methods typically include tasks such as data entry, assembly line operations, and basic rule-based processes. Traditional automation relies heavily on predefined rules and structured data. Intelligent automation incorporates technologies such as AI, ML, NLP, and RPA to perform more complex tasks that traditionally require human cognition [9, 10]. Intelligent automation systems can analyze unstructured data, make decisions, learn from experience, and adapt to changing circumstances. They often involve cognitive capabilities such as problem solving, decision making, and language understanding.

The evolution from traditional automation to intelligent automation involves several key advancements. Traditional automation operates based on predetermined rules, while intelligent automation can make decisions by analyzing data, identifying patterns, and understanding context. This shift allows for greater flexibility and adaptability in managing diverse scenarios. A crucial component of intelligent automation is the integration of Artificial Intelligence (AI) and Machine Learning (ML) algorithms, which enable continuous improvement of processes. These algorithms learn from data inputs and user interactions, enhancing the efficiency and effectiveness of systems over time [11]. Additionally, Natural Language Processing (NLP) is used to enable systems to understand and process human language, which facilitates interactions through chatbots, virtual assistants, and voice commands. This ability significantly expands the scope of tasks that can be automated and enhances user experience. Furthermore, intelligent automation goes beyond handling repetitive tasks to perform more complex activities that require cognitive capabilities. This includes tasks like data analysis, anomaly detection, and decision-making in situations involving uncertainty or incomplete information [12]. Advanced robotics, a core aspect of intelligent automation, empowers robots to carry out tasks traditionally performed by humans. These systems can learn from human demonstrations, adapt to new environments, and work alongside human workers in shared spaces.

Intelligent automation, particularly through AI, enables personalized learning experiences by adapting educational content to individual student needs, preferences, and abilities. This customization fosters self-directed learning and promotes student autonomy [13–15]. AI-driven systems can provide real-time feedback and adaptive assessments, which help in identifying students’ strengths and weaknesses, allowing for timely interventions and support [14, 16]. AI applications such as intelligent tutoring systems, virtual classrooms, and adaptive learning platforms have shown to significantly improve student engagement and learning outcomes. These technologies create interactive and immersive learning environments that keep students motivated and involved [15, 17, 18]. Studies have demonstrated that AI has a small to medium positive effect on students’ learning achievements, indicating its potential to enhance educational outcomes across various disciplines and educational stages [16].

1.1 Overview of RPA architecture and components

RPA is transforming industries by automating repetitive, rule-based tasks, enhancing operational efficiency, and reducing human error. RPA systems typically follow a layered architecture shown in Fig. 1 that involves multiple components, each playing a vital role in executing automation tasks. At the heart of RPA is the Bot Creation/Design Layer, which consists of tools that developers use to design and configure automation workflows. Popular platforms such as UiPath, Automation Anywhere, and Blue Prism offer intuitive drag-and-drop interfaces for defining actions for bots. The Bot Execution Layer comprises software robots that run on end-user machines or virtual environments. These bots replicate human actions, such as keyboard inputs and mouse movements, interacting with software interfaces according to predefined rules and conditions. Centralized control of bots is ensured by the Bot Control/Orchestration Layer, which manages monitoring, scheduling, and bot lifecycle functions. Tools like UiPath Orchestrator

Fig. 1 Basic RPA Architecture

and Automation Anywhere's Control Room allow administrators to distribute workloads, manage bot resources, and maintain logs. Since RPA often handles sensitive data, the Security Layer is crucial, providing secure credential management, role-based access control, and encrypted data storage, with integrations like CyberArk and Azure Key Vault to enhance security.

RPA bots interact with applications at both the user interface (UI) and application programming interface (API) levels. At the UI level, bots perform tasks by interacting with the interface of existing systems, using methods like image recognition and keystroke simulation. The major advantage is that no backend integration is required, making it suitable for automating legacy systems. However, UI changes, such as shifts in layout or design, may break the automation and necessitate bot reconfiguration. For more efficient and reliable automation, bots can interact with systems through APIs, which bypass the front-end UI, offering faster execution times and reducing maintenance efforts as they are unaffected by UI changes. To automate business processes, RPA platforms rely on core components like Robots, Recorders, Workflow Designers, Object Repositories, and AI and Cognitive Components. Robots, which can be either attended or unattended, are the primary agents of automation, responsible for executing tasks like data entry or invoice processing. RPA tools also offer recording capabilities that allow users to capture and replay human interactions with software interfaces. The Workflow Designer provides a graphical interface for users to design, test, and debug automation workflows, enabling the creation of sequences, loops, and decision-making logic. The Object Repository serves as a centralized storage location for application objects that bots interact with, ensuring reusability across tasks. With the integration of artificial intelligence (AI) and machine learning (ML) capabilities, RPA systems can handle unstructured data, natural language processing (NLP), and decision-making tasks, such as sentiment analysis, optical character recognition (OCR), and predictive analytics.

Several key technologies enable RPA, allowing automation across different software platforms and environments. Screen scraping, one of the earliest technologies used in RPA, enables bots to capture visual data from screens and extract text or images for input into other systems. Although modern RPA platforms have improved this technique, it remains limited in handling dynamic interfaces. API integrations, on the other hand, offer a faster and more reliable method of connecting systems. Bots can use APIs to interact with third-party applications, retrieve data from databases, or submit forms through web services. Optical Character Recognition (OCR) is particularly useful in document-heavy industries, enabling bots to extract text from scanned images, PDFs, and other documents. RPA platforms like UiPath integrate third-party OCR tools to enhance text extraction accuracy. NLP enables RPA systems to interpret and process human language, which is essential for automating tasks involving chatbots, emails, or voice assistants. By parsing natural language, bots can understand user queries, generate responses, or initiate processes. The inclusion of Machine Learning (ML) algorithms allows RPA bots to evolve from rule-based systems into more intelligent and adaptive solutions, optimizing workflows by learning from past interactions and identifying patterns in data.

2 Literature review

In this section, a literature review synthesizes the current research on the application of RPA in education, examining its impact on operational efficiency, pedagogical practices, and the overall learning environment.

Gunawan and Wijaya explored the use of RPA to automate attendance checking and reporting processes in universities, demonstrating a 99.9% reduction in time spent on these tasks, along with a 43% reduction in costs over five years. This study underscores how RPA can significantly enhance efficiency and productivity within administrative departments, enabling staff to focus on higher-value tasks [19]. Similarly, Wijaya and Gusssssnawan examined RPA's potential in automating students' internship report checking and recapitulation, with the results showing a 99.9%-time reduction and a 67% decrease in associated costs. These findings highlight RPA's role in increasing administrative productivity while reducing costs in educational institutions [20]. Beyond administrative tasks, RPA has also been utilized to support teaching activities. Nakić and Boban introduced an innovative approach to automating course searches on e-learning platforms, which allows students and teachers to quickly find relevant courses, thereby improving the overall learning experience. Their algorithm interacts with platform interfaces, extracts necessary data, and generates structured reports, saving time and ensuring greater accuracy in the process [21]. In a similar vein, Phakdee et al. integrated RPA with Acceptance Test-Driven Development (ATDD) to automate teaching support tasks, such as checking word errors in student reports. Their study found a 71.42-fold reduction in time spent on error checking, enabling educators to dedicate more time to valuable instructional activities [22]. RPA's impact extends to the examination process as well. Khan and Tailor explored how RPA could automate the handling and processing of student examination results. Their research showed that RPA could replace manual procedures, such as converting marksheets from PDF to Excel format, thus reducing the risk of human error and the time spent on data entry. This process improvement could significantly change how universities manage student assessments, making the examination process more accurate and efficient [23].

Yung-Hsiang Hu et al. in [24] explored the integration of RPA and predictive analytics to develop an Intelligent Tutoring Robot (ITR) for distance learning students. To compare the interactions between the ITR and students with those between human teachers and students, the study used a randomized control group pretest–posttest methodology. The control and experimental groups did not significantly differ in terms of academic performance; however, the experimental group received better average grades. Corneliu Octavian Turcu and Cristina Turcu's has examined how RPA can enhance digitalization in higher education by addressing relevant tasks and challenges. A case study demonstrated RPA's ability to handle basic preprocessing tasks but emphasized the importance of strategic planning and platform selection. This discussion highlights the need for RPA to evolve toward handling diverse tasks and unstructured data, signaling the emergence of AI-enhanced "smart" robots for higher education applications [25]. Monica-Ioana Vulpe and Vladimir-Aurelian Enăchescu's work explores the impact of technology, particularly robotic process automation (RPA), on education and child development. It emphasizes society's collective responsibility for a child's education and highlights the transformative influence of technology on modern life. The lens of RPA suggests practical solutions for managing various aspects of adult life that can be translated into educational contexts for children [26].

M. Lebens et al. investigates the relevance of teaching robotic process automation (RPA) platforms from the Gartner Leaders Quadrant in higher education to enhance students' career readiness [27]. By analyzing current job postings, it is concluded that incorporating these platforms into the curriculum is beneficial for students seeking careers in the RPA sector. The findings provide valuable insights for faculty to align teaching with industry demands, facilitating students' entry into the rapidly growing field of RPA technology. N. R. Mosteanu examines the urgent need for machine learning and digitization in higher education accelerated by the COVID-19 pandemic. Innovative applications of artificial intelligence, such as physical robots for campus guidance and robotic process automation for admission and evaluation processes, are proposed to enhance teaching effectiveness and streamline administrative tasks in virtual learning environments [28]. Hee-Kyung Hong suggests leveraging web data for RPA learning and presents a model for automated learning, focusing on rule-based decision logic [29]. It outlines six steps for designing learning using Brity RPA and proposes a case study using the 'Piano Tiles' game to facilitate empirical studies on automation implementation.

Table 1 provides a concise overview of the focus, methodology, key findings, and implications of each study, highlighting the diverse applications and considerations of RPA in educational contexts.

Table 1 Summary of key research on RPA in higher education and administrative processes

Author(s)	Research Focus	Key Findings/Contributions
Gunawan & Wijaya	Automation in administrative processes	Demonstrated a 99.9% reduction in time for attendance checking and a 43% reduction in costs over five years, highlighting RPAs ability to enhance administrative productivity [19]
Wijaya & Gunawan	Automation in student internship report management	Achieved a 99.9%-time reduction and 67% cost reduction in managing internship reports, emphasizing RPA's role in administrative efficiency [20]
Nakić & Boban	Course search automation in e-learning platforms	Proposed an RPA solution that automates course searches, improving accuracy and saving time for students and teachers in the learning process [21]
Phakdee et al.	RPA for automating teaching support tasks	Integrated RPA with ATDD to reduce the time spent on error checking in student reports by 71.42-fold, enabling educators to focus on more meaningful tasks [22]
Khan & Tailor	RPA for automating student examination result processing	Showed that RPA can automate the conversion of marksheet from PDF to Excel, reducing errors and time spent on data entry in examination processes [23]
Yung-Hsiang Hu et al.	RPA in Intelligent Tutoring Robots (ITR) for distance learning	Found that students interacting with the ITR achieved better grades than those with human teachers, though academic performance between groups did not significantly differ [24]
Corneliu Octavian Turcu & Cristina Turcu	RPAs role in digitalization in higher education	Examined RPA's ability to handle preprocessing tasks and discussed the importance of strategic planning and platform selection for effective RPA integration [25]
Monica-Ioana Vulpe & Vladimir-Aurelian Enăchescu	Technology, RPA, and child development	Focused on the transformative influence of RPA and technology in education, emphasizing collective responsibility for children's education [26]
M. Lebens et al.	Incorporating RPA platforms into curricula	Recommended teaching platforms from the Gartner Leaders Quadrant to enhance career readiness for students in the RPA industry, linking education to industry demands [27]
N. R. Mosteanu	AI and RPA in higher education	Suggested the use of RPA for admissions and evaluation processes, and physical robots for campus guidance to enhance teaching effectiveness [28]
Hee-Kyung Hong	RPA in automated learning	Proposed a model for automated learning using RPA, focusing on rule-based decision logic and presented a case study using the 'Piano Tiles' game for empirical studies [29]

2.1 RPA’s mixed impact on pedagogy

The application of RPA in pedagogy has shown both positive and mixed outcomes, with notable improvements in some areas and limitations in others. Studies such as those by Nakić and Boban [21] and Phakdee et al. [22] have demonstrated how RPA can streamline teaching support tasks, such as course searches and error checking in student reports. These applications free up educators’ time, allowing them to focus on more engaging instructional activities. However, the impact of RPA on direct teaching effectiveness remains debated. Yung-Hsiang Hu et al.[24] found that while ITRs enhanced some aspects of student learning, they did not significantly outperform human teachers in terms of academic achievement, even though students interacting with ITRs received better grades on average.

This contrast suggests that while RPA can support learning through automation, it may not fully replicate the personalized and adaptive nature of human teaching. Additionally, as RPA continues to evolve, its potential to support more complex pedagogical tasks remains uncertain. RPA’s mixed impact thus reflects the challenge of balancing automation with the need for human interaction in education, indicating that while RPA can enhance operational efficiency, its role in fostering deep, meaningful learning remains limited.

3 Methodology design

This section outlines the systematic approach followed in conducting the literature review on Robotic Process Automation (RPA) in education from 2020 to 2024. The methodology adheres to established guidelines for conducting systematic literature reviews (SLRs) to ensure a comprehensive, transparent, and replicable review process.

3.1 Research questions

The review is structured around four key research questions (RQs) designed to explore RPA’s applications and implications in higher education. The Table 2 outlines the primary research questions addressing RPA adoption, success factors, benefits, challenges, and ethical considerations in educational settings."

3.2 Search strategy

To ensure a comprehensive literature search, thze following academic databases were selected: IEEE Xplore, Scopus, Web of Science, and Google Scholar. The search strategy involved developing structured keyword combinations, focusing on the intersection of RPA and education. The search keywords included:

- "Robotic Process Automation" AND "education"
- "RPA" AND "teaching" OR "learning"
- "Process automation" AND "higher education"
- "Automation in education"

Table 2 Key research questions (RQs) exploring the applications and implications of RPA in higher education

Research Question (RQ)	Description
RQ1	How has Robotic Process Automation been adopted and utilized across different educational institutions and processes?
RQ2	What are the common success factors and challenges identified in case studies of RPA adoption in higher education institutions?
RQ3	What are the key benefits of RPA implementation in the education industry, and how has it enhanced efficiency in administrative and teaching tasks, as well as student-centered learning experiences?
RQ4	What challenges and ethical considerations have arisen from RPA adoption in education, and how are institutions addressing these issues?

The inclusion period was limited to papers published between 2020 and 2024, ensuring that only the most recent and relevant studies were considered. An example of a search string is: ("RoboticProcessAutomation"OR"RPA") AND ("education"OR"teaching"OR"learning") AND("2020"OR"2021"OR"2022"OR"2023"OR"2024").

3.3 Inclusion and exclusion criteria

Table 3 outlines the criteria used to select relevant studies, ensuring a focus on RPA applications in education. It details the specific inclusion and exclusion requirements, emphasizing peer-reviewed studies published between 2020 and 2024 and excluding materials unrelated to software-driven RPA in the educational sector.

3.4 Quality assessment

Table 4 outlines the criteria used to assess the quality of each selected paper, focusing on relevance, methodological rigor, contribution to knowledge, and the quality of reporting and transparency.

SLR followed the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines to ensure a transparent and replicable selection process. Below Fig. 2 shows the detailed account of the stages, along with the respective number of records at each step:

- 1 *Identification* A total of 120 records were identified through searches in academic databases, including IEEE Xplore, Scopus, Web of Science, and Google Scholar. These records were sourced using a combination of keywords related to Robotic Process Automation (RPA) in education and covered publications from 2020 to 2024.
- 2 *Screening* After removing duplicates, 104 unique records remained. These records were screened by reviewing their titles and abstracts to determine their relevance to the research questions. Studies that did not focus on RPA in education or were outside the inclusion period were excluded during this stage.
- 3 *Eligibility* The full texts of 62 articles were further assessed for eligibility based on predefined inclusion and exclusion criteria. This step involved an in-depth review of each article to ensure the content addressed the applications of RPA in educational settings.
- 4 *Inclusion* Following the eligibility assessment, 54 studies were included in the final review from years 2020 to 2024 shown in Fig. 3. These studies were used for data extraction and synthesis to answer the research questions of the review.

4 Results and discussion

4.1 How has RPA been adopted and utilized across different educational institutions and processes?

Robotic Process Automation (RPA) has been increasingly adopted across various educational institutions and processes, reflecting its transformative potential in enhancing operational efficiency and streamlining administrative tasks. For

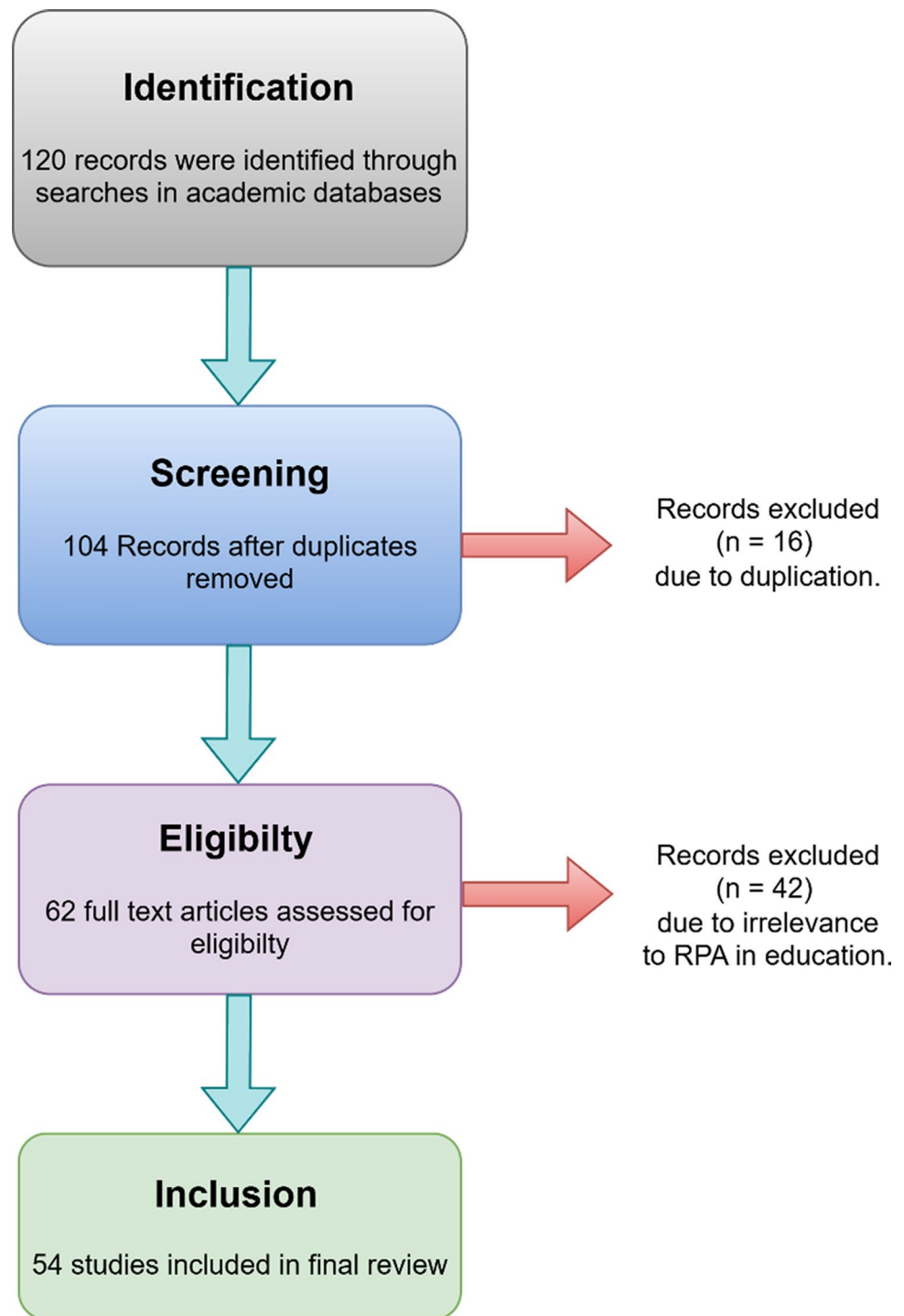
Table 3 Inclusion and exclusion criteria for study selection

Criteria Type	Details
Inclusion Criteria	• Papers published between 2020 and 2024 • Peer-reviewed research articles, conference papers, and journal articles • Studies focusing on the application of RPA in education • Papers published in English
Exclusion Criteria	• Studies outside the educational sector • Non-peer-reviewed materials • Papers focusing primarily on hardwired robots to ensure a focus on software-driven automation • Articles using 'robotics' without specific emphasis on RPA • Duplicate articles across multiple digital platforms

Table 4 Quality assessment criteria for evaluating selected studies

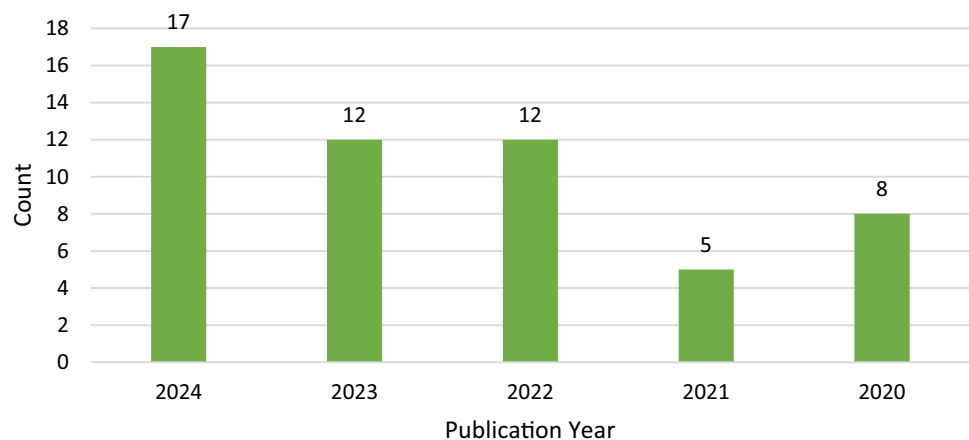
Assessment Criteria	Questions for Evaluation	Key Considerations
Relevance	Does the paper contribute to answering the research questions?	Does the paper address core topics/issues in the research question? Does it cover necessary breadth and depth? Is the paper timely and reflect current trends?
Methodological Rigor	Is the research methodology robust and well-explained, including research design, data collection, and analysis techniques?	How applicable are the findings to the research context? Is the research design clearly stated and suitable? Are the data collection methods appropriate and described in detail? Are the data analysis techniques explained and relevant? Does the study control for biases and ensure validity and reliability? Is the study replicable with clear transparency in the reporting of methodology and procedures?
Contribution to Knowledge	Does the paper offer new insights or innovations within the field?	Does the paper present novel ideas, frameworks, or findings? How significant are the findings for advancing knowledge? Does it address gaps in the existing literature or field?
Quality of Reporting and Transparency	How transparent is the reporting of methods, findings, and interpretations?	Are actionable implications for practice or policy provided? Is the methodology and results clearly communicated? Does the paper omit any critical details that might hinder replicability? Is there transparency in the research process (e.g., data sharing, ethical considerations)?

Fig. 2 Stages of Record Selection in the Systematic Literature Review (SLR)



instance, academic librarians in Hong Kong have demonstrated a willingness to learn about and implement RPA in their libraries, indicating a positive attitude toward adopting innovative technologies despite their initial unfamiliarity with the concept. This openness signifies significant potential for enhancing library services [30]. Additionally, universities are utilizing RPA to streamline student club enrollment processes through tools like UiPath. This implementation not only enhances operational efficiency but also fosters greater student engagement and satisfaction [31]. Moreover, research focusing on vocational college accounting courses reveals that the perceived usefulness and ease of use of RPA significantly influence its integration into accounting education, demonstrating the technology's adaptability in specialized fields [32]. Furthermore, a private university adopted RPA to automate the checking and recapitulation of students'

Fig. 3 Number of studies included from 2020 to 2024



internship reports, which has led to significant time and cost savings compared to manual processes. This application underscores the potential of RPA to alleviate administrative burdens, allowing educational staff to concentrate on more strategic initiatives [20, 33].

Additionally, RPA has been explored as a transformative tool across various industries, including education, highlighting its role in streamlining processes and enhancing operational efficiency [34]. Specific applications of RPA in education include the automation of attendance checking and reporting in private universities, which has been shown to significantly reduce time and costs associated with traditional methods [19, 35]. At İzmir Bakırçay University, RPA has been implemented to digitize and automate back-office functions, enhancing user experience for students and improving institutional operations [36]. Moreover, there is a growing advocacy for incorporating RPA tools into higher education curricula to boost students' career readiness. By focusing on tools that are increasingly recognized in job postings, educational institutions can better prepare students for the workforce [27]. Additionally, the adoption of RPA is viewed as a strategic tool to enhance administrative efficiency in higher education institutions, with a theoretical model suggesting that RPA can mediate Business Process Management (BPM) to support digital transformation efforts [37]. In the context of student management systems, RPA can automate repetitive tasks and improve data management, providing accessible web-based solutions across various platforms [38].

Finally, while RPA is gradually gaining traction within educational settings, many institutions still exhibit hesitation in adopting such technologies, highlighting the need for further exploration of successful use cases to alleviate bureaucratic burdens [39]. Collectively, these examples underscore the diverse applications of RPA in educational contexts, illustrating its potential to optimize processes, improve institutional effectiveness, and contribute to a more responsive and adaptive educational landscape. RPA is increasingly being adopted in various sectors, including education. RPA uses software robots or "bots" to automate repetitive, rule-based tasks traditionally performed by humans. In the context of education, RPA can streamline administrative processes, enhance the efficiency of operations, and improve the overall experience of students, educators, and administrative staff [40, 41]. The following are some specific applications of RPA in education:

Administrative Processes Automation: One of the primary areas where RPA can have a significant impact on education is automating administrative tasks. Educational institutions, ranging from schools to universities, address a plethora of administrative processes, including admissions, enrollment, student records management, and payroll processing. By implementing RPA solutions, institutions can streamline these processes, reduce manual errors, enhance efficiency, and save valuable time and resources [42]. **Student Support Services:** RPA can also play a crucial role in enhancing student support services. Chatbots powered by RPA technology can provide 24/7 assistance to students, addressing queries related to admissions, course registrations, academic schedules, etc.etc. These virtual assistants can offer personalized guidance to students, helping them navigate through various academic and administrative challenges [38]. **Budgeting and financial planning** are critical for educational institutions. RPA can significantly contribute by automating tasks such as expense tracking, payroll management, and accounts payable. Furthermore, RPA can help identify cost-saving opportunities and provide valuable insights into financial trends, enabling administrators to make informed decisions [43]. Educational institutions must adhere to various regulations and reporting requirements. RPA can ensure compliance by automating data collection and reporting processes, as well as maintaining audit trails and clear records of compliance-related actions [44]. Beyond administrative tasks, RPA has the potential to enhance teaching and learning experiences. Intelligent tutoring systems powered by RPA can adjust to the unique learning needs of students, offering personalized

learning paths and feedback. Additionally, RPA tools can automate grading and assessment, freeing educators to focus more on teaching and engaging with students.

4.2 What are the common success factors and challenges identified in case studies of RPA adoption in higher education institutions?

In the context of Robotic Process Automation (RPA) adoption in higher education institutions, several common success factors and challenges have been identified through various case studies. Key success factors influencing RPA adoption in vocational colleges include perceived usefulness, ease of use, social influence, and facilitating conditions, which collectively shape the willingness of institutions to integrate this technology into their operations [32]. However, challenges often arise during implementation due to practical considerations and a lack of relevant education among accountants. This underscores the need for educational institutions to adequately prepare for the integration of RPA, ensuring that stakeholders possess the necessary skills and knowledge to utilize the technology effectively [45].

The COVID-19 pandemic has further accelerated the need for digital transformation in higher education, creating additional challenges that necessitate the exploration of affordable digital solutions, including RPA. Institutions must navigate these complexities to ensure a smooth transition to digital platforms while maintaining operational continuity [25]. The development of an early-warning system through RPA illustrates the potential of predictive analytics in supporting distance learning; interestingly, studies show no significant performance difference between intelligent tutoring robots and human teachers, indicating that RPA can effectively complement traditional teaching methods [24]. Moreover, as education evolves through the transition between classicism and modernism, RPA plays a vital role in addressing contemporary challenges. However, resistance to adopting new technologies remains a significant barrier, highlighting the societal responsibilities of educational institutions in facilitating this transition [26]. Training for teachers and staff is crucial for effective RPA implementation, as it addresses challenges related to potential resistance to technology and emphasizes the importance of social and emotional intelligence in robotic interfaces [40].

Additionally, an automatic analytics model for learning skills analysis using game player data and RPA in serious educational games showcases success factors such as data-driven approaches and engagement. However, ensuring the accuracy and relevance of the data analysis presents its own set of challenges [46]. Finally, the integration of RPA in higher education is often hampered by limited budgets and human resources, emphasizing the need for institutional support and adequate infrastructure to overcome these challenges [39]. Collectively, these insights reveal that while RPA adoption offers significant opportunities for enhancing educational processes, addressing the identified challenges is critical for successful implementation and sustained impact.

Universities worldwide are leveraging robotic process automation (RPA) to enhance operational efficiency by automating repetitive administrative tasks. This includes streamlining admission processes, managing student services, and handling human resources and financial operations. By integrating RPA, universities can reduce errors, accelerate processing times, and allow staff to focus on more strategic initiatives, ultimately improving the overall educational experience. The following are a few case studies demonstrating the impact of RPA in higher education:

4.2.1 Case study 1: University of Melbourne's automation success

The University of Melbourne has successfully leveraged the Automation Success Platform to enhance administrative operations, particularly in student admissions, faculty administration, and supplier tracking. By automating repetitive tasks, the university has significantly improved efficiency, reduced manual labor, and created a more streamlined experience for both staff and students [47]. The university has automated 22 processes, leading to a 97% improvement in supplier detail processing and saving 10,000 h of labor annually. These efforts have not only reduced time spent on manual tasks but also allowed staff to focus on more strategic and value-added activities.

In terms of operational impact, automation has brought significant improvements to key administrative areas. Student admissions have been expedited by automating data entry, making the process faster and less prone to errors, while improving the experience for prospective students. Faculty administration has also been streamlined, relieving faculty members from repetitive administrative duties and allowing them to focus on teaching and research. Finally, the automation of supplier tracking has led to more accurate and efficient handling of supplier details, enhancing operational workflows and contributing to better resource allocation. The University of Melbourne's adoption of the Automation Success Platform demonstrates how RPA can transform university operations, leading to increased efficiency, labor savings, and improved quality of education.

4.2.2 Case study 2: RPA at the University of Auckland

The University of Auckland has implemented Robotic Process Automation (RPA) across various administrative and academic processes, improving operational efficiency and enhancing both staff and student experiences [48]. Serving over 40,000 students and with the support of 5,500 staff members, the university has saved approximately 23,000 h annually through RPA. Automation has led to impressive results, with 96.2% of orchestration runs being completed successfully and 99% success rate in finance, ensuring high accuracy in financial operations. Additionally, 98% client satisfaction in finance indicates that automated processes not only perform well but also meet the needs of users effectively.

The integration of RPA has freed staff from mundane tasks, allowing them to focus on more human-centered services and enhancing student experiences. The university has also incorporated automation into its educational offerings, launching an RPA course and Automation Youth course to prepare students for future work opportunities in technology and automation. Despite initial challenges with process maturity and documentation, the university optimized its operations before deploying UiPath robots. This has led to the automation of processes such as student transcript generation and supply setups, significantly reducing turnaround times and enhancing operational efficiency. Furthermore, the university's partnership with UiPath to create an RPA course exemplifies its commitment to integrating automation into its academic curriculum. Although there was initial skepticism among staff, they have now fully embraced the technology, recognizing its benefits in improving internal operations and overall customer satisfaction. Looking ahead, the university plans to expand its automation efforts, incorporating machine learning and sharing its knowledge with the broader community to foster innovation and education in automation.

These case studies from the University of Melbourne and the University of Auckland highlight the transformative impact of RPA in higher education, demonstrating how automation can improve operational efficiency, reduce manual labor, and provide better learning experiences for students. Institutions across the globe are increasingly adopting robotic process automation (RPA) to streamline various administrative functions critical to their operations. Figure 4 shows an overview of how universities are implementing RPA to enhance efficiency [49].

These universities leverage RPA to optimize admissions processes, manage faculty affairs efficiently, enhance student recruitment efforts, streamline financial aid processing, automate transcript generation, and maintain comprehensive student records. RPA adoption in higher education aims to improve operational efficiency and provide better services to students and faculty alike. Figure 4 illustrates the number of universities using Robotic Process Automation (RPA) for different administrative tasks. From the figure it is clear that which tasks have seen wider RPA adoption across various universities. For instance, tasks like Student Recruitment and Transcript Generation are automated by multiple universities, reflecting a higher number of institutions that use RPA in these areas. In contrast, tasks such as Faculty Administration show limited RPA adoption, with fewer universities leveraging automation for these administrative functions. This visualization provides a clear view of the spread of RPA usage across higher education institutions. It highlights areas where universities are most likely to implement RPA to streamline administrative operations, and shows where RPA adoption is less widespread, which may suggest potential opportunities for future implementation in educational institutions.

The Fig. 5 provides a comparative analysis of the specific administrative tasks where different universities have adopted Robotic Process Automation (RPA). It highlights which universities use RPA for tasks such as admissions processing, faculty administration, student recruitment, financial aid processing, transcript generation, and student records management. For example, the University of Melbourne utilizes RPA for admissions processing, faculty administration, and student recruitment, while other universities, like the University of Texas at Austin, focus on student recruitment and financial aid processing.

4.3 What are the key benefits of RPA implementation in the education industry, and how has it enhanced efficiency in administrative and teaching tasks, as well as student-centered learning experiences?"

The implementation of Robotic Process Automation (RPA) in the education industry has yielded numerous key benefits, particularly in improving efficiency in both administrative and teaching tasks. One of the most significant efficiency gains has been observed in the administrative tasks associated with student internship reporting processes, where RPA achieved an astonishing 99.9% reduction in time spent, highlighting its transformative potential for streamlining operations [20]. Similarly, the integration of RPA into student club enrollment processes not only

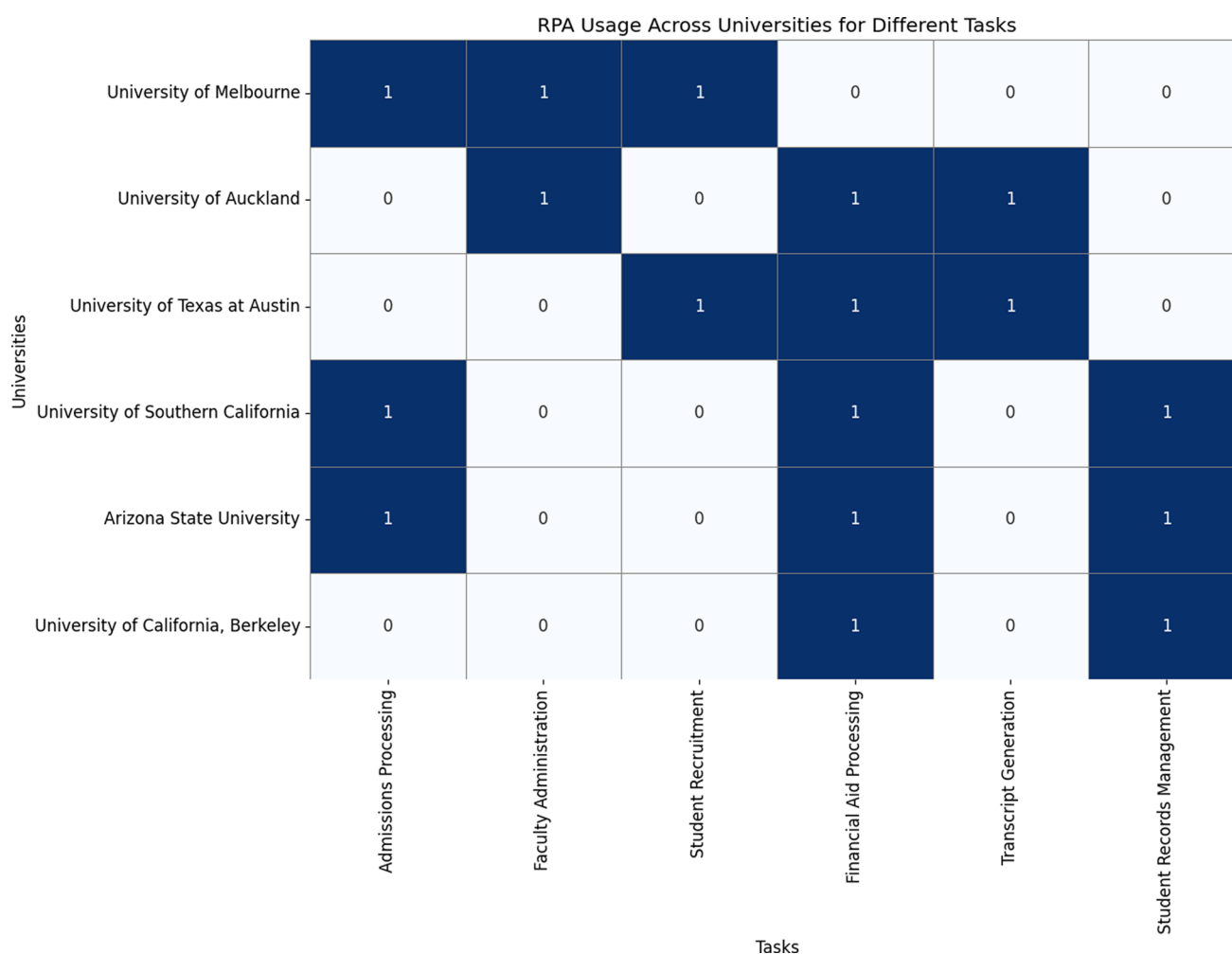


Fig. 4 Comparative analysis of RPA implementation in higher education institutions

enhanced administrative efficiency but also aimed to reduce ecological footprints, showcasing RPA's ability to create a more streamlined and sustainable administrative environment [31].

Furthermore, RPA technology has the potential to fundamentally transform educational methodologies by improving student engagement and providing timely feedback, thereby enhancing overall academic performance [6]. In attendance checking and reporting, RPA facilitated a remarkable 99.9% reduction in time spent on these processes, accompanied by a 43% cost reduction over five years, further underscoring the substantial efficiency gains associated with its application [19]. Additionally, the digitization and automation of university processes with RPA resulted in a significant 96.97% time saving in processing tasks, contributing to enhanced administrative efficiency [36]. Figure 6 depicts how RPA influences different educational functions, from administrative tasks to personalized learning support.

RPA's capabilities extend beyond mere time savings; it offers benefits in managing large volumes of data, improving accuracy, and reducing workload across various sectors, including education [7]. Moreover, the implementation of RPA facilitates digital transformation in higher education institutions, allowing them to focus on core educational activities rather than repetitive administrative tasks, thereby optimizing resource allocation and enhancing productivity [37]. The training of educators also benefits from RPA, as it provides personalized and efficient training solutions that improve performance and adaptability to new technologies [40]. Finally, by automating student management systems using chatbot and RPA technology, institutions can efficiently manage student data and notifications, freeing faculty and administrators from tedious tasks and significantly enhancing the overall educational experience for students [38]. Collectively, these insights affirm that RPA not only streamlines administrative processes but also enhances the teaching and learning environment, positioning educational institutions for greater efficiency and effectiveness in their operations.

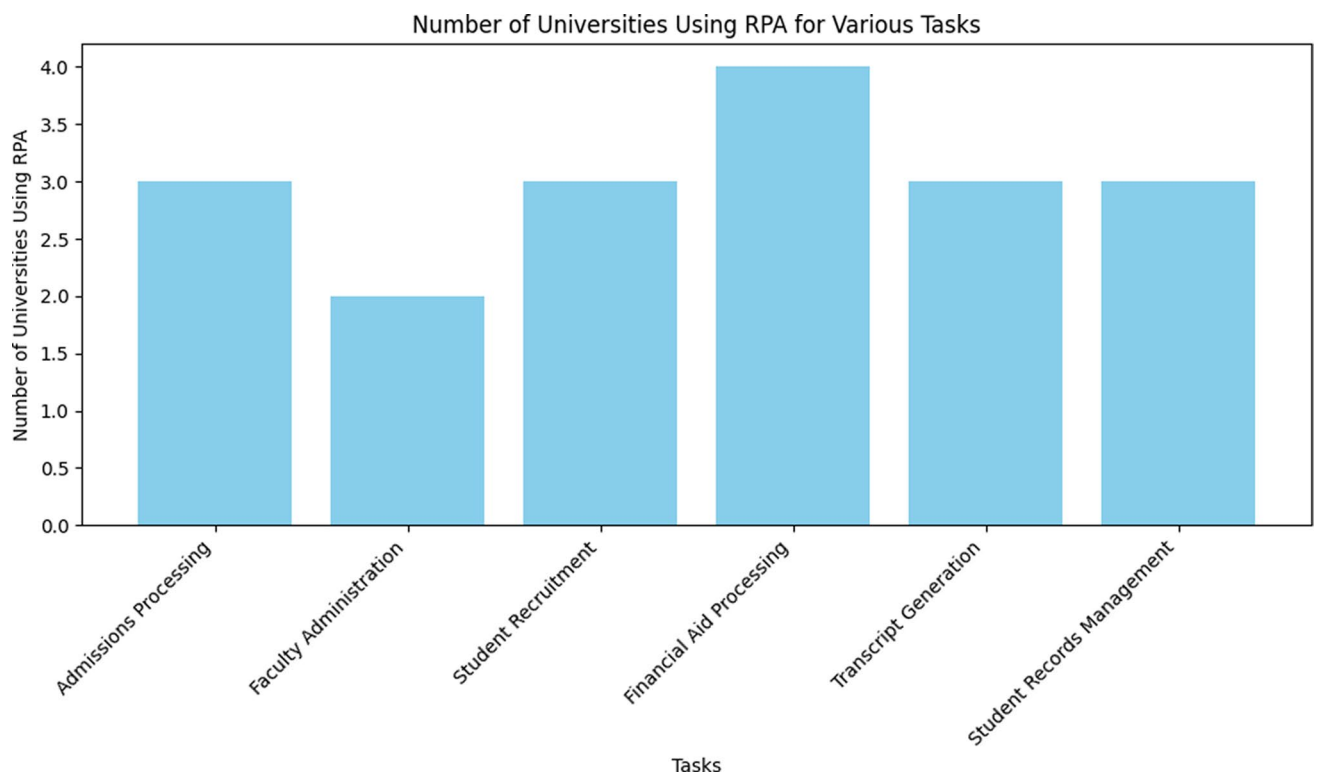


Fig. 5 Number of universities using RPA for different Tasks

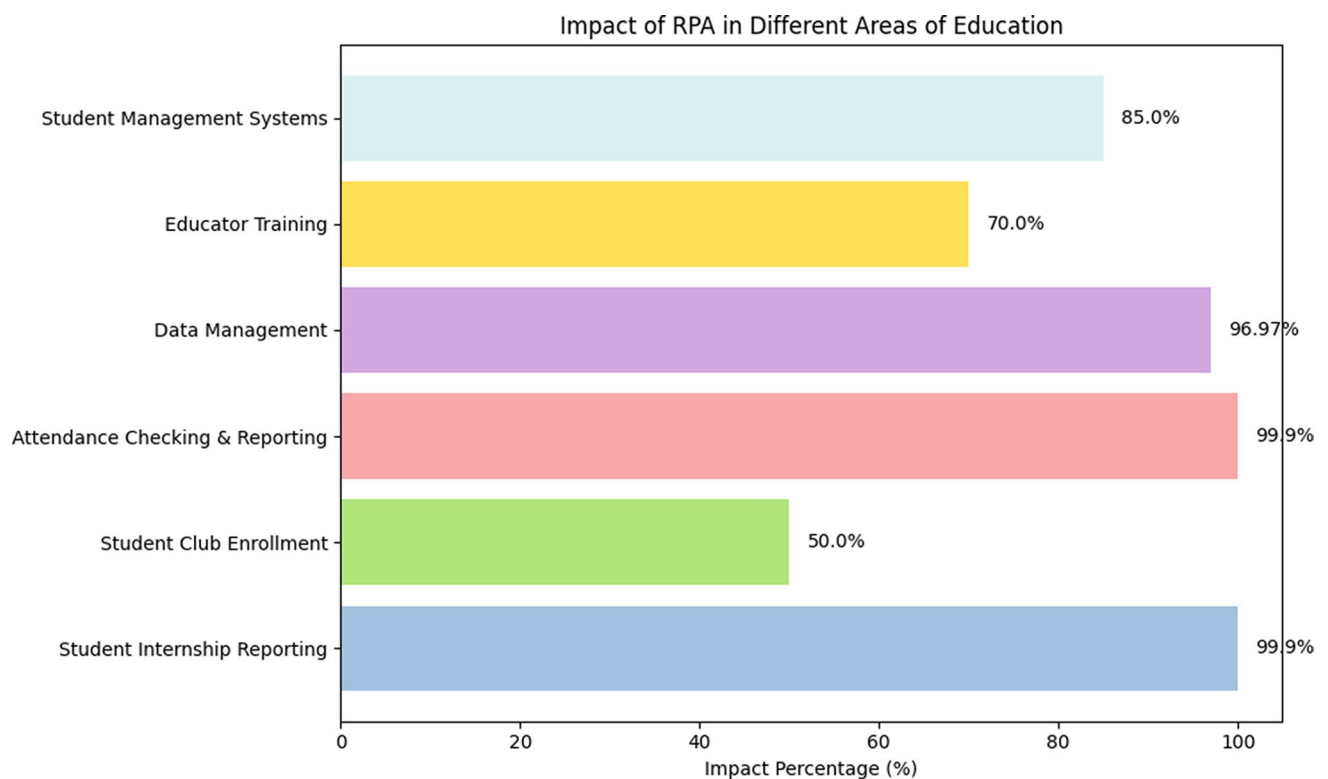


Fig. 6 Impact of RPA Across Various Educational Functions

RPA has significant potential to enhance student-centered learning by automating repetitive tasks, allowing educators to focus more on personalized instruction and student development shown in Fig. 7. RPA can automate administrative tasks such as course searches on e-learning platforms, data collection, and report generation, which traditionally consume a lot of time for educators and students [21, 22, 50, 51]. For instance, RPA can streamline the process of recording student grades, reducing errors and saving time for teachers [52]. Additionally, RPA can be integrated into Intelligent Tutoring Systems (ITS) to provide automated responses and personalized feedback to students, supporting distance learning [24]. Automated reporting tools can generate detailed performance reports for students, helping mentors provide targeted guidance and support [53]. Furthermore, RPA can be used to develop interactive learning tools, such as quiz-based modules integrated with humanoid robots, which can enhance student engagement and provide immediate performance evaluations [54].

4.3.1 Impact of RPA in the education industry

The adoption of Robotic Process Automation (RPA) is significantly transforming the education sector, providing universities and educational institutions with the tools to enhance operational efficiency, reduce costs, and improve service delivery. RPA's ability to automate repetitive, rule-based tasks has made it a valuable asset for institutions looking to streamline their administrative processes. The following points highlight the notable impacts and statistics regarding the adoption of RPA in the education industry.

According to Gartner, by 2024, 40% of educational institutions are expected to be utilizing RPA. This indicates a rapid shift toward automation in higher education, as institutions seek to improve operational efficiency and focus more on student-centric activities. In a related survey by UiPath, 75% of educational institutions have already adopted RPA, with an additional 20% planning to implement RPA in the next year. This widespread adoption reflects the growing recognition of RPA's benefits and the increasing demand for automation in education. RPA is anticipated to generate significant cost savings for the education sector. The sector is projected to save approximately \$2.5 billion annually by 2025 through the implementation of RPA. This savings is primarily due to the reduction in manual labor, which allows educational institutions to reallocate resources more effectively and focus on core educational objectives rather than administrative tasks. RPA has the potential to automate up to 70% of repetitive tasks in the education sector. These tasks include admissions processing, student records management, financial aid processing, and transcript generation. By automating these processes, educational institutions can reduce human error, speed up operations, and free up staff to focus on more complex and strategic activities that add greater value to the

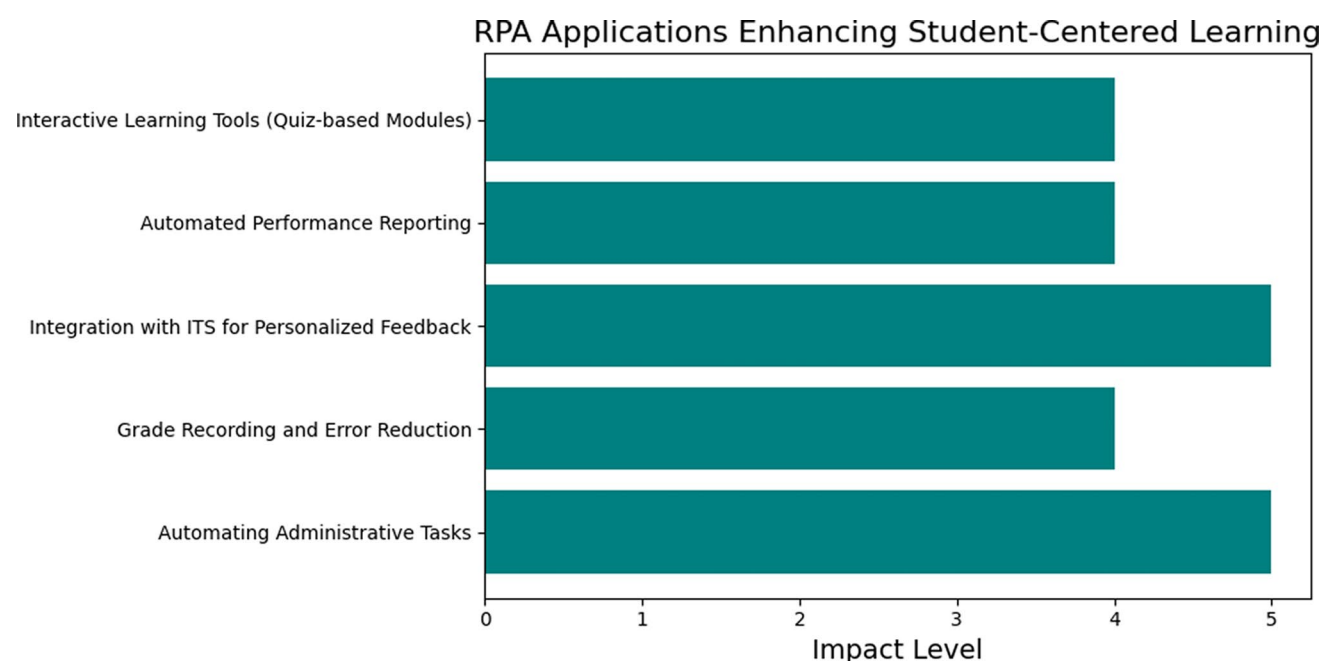


Fig. 7 Role of RPA in Enhancing student-centered learning

institution and its students. Figure 8 illustrates the adoption rates of RPA in education and the potential cost savings associated with its implementation.

These statistics underscore the transformative impact of RPA on the education industry, driving advancements in administrative processes, student services, and resource management.

4.3.2 Benefits of RPA in education

Implementing robotic process automation (RPA) in the education sector offers several significant advantages shown in Fig. 9.

- **Enhanced Efficiency and Productivity:** Automating repetitive tasks with RPA significantly boosts the efficiency and productivity of educational institutions, allowing staff to focus more on strategic projects and student-centered activities.
- **Cost Savings:** Implementing RPA for administrative processes can lead to substantial cost savings by decreasing the need for manual labor and reducing errors from manual data entry.
- **Improved Accuracy and Compliance:** RPA systems ensure consistent and accurate data processing while adhering to regulatory requirements, thus minimizing the risks of errors and noncompliance.
- **Personalized Learning Experiences:** RPA-powered educational tools can offer personalized learning experiences by delivering tailored content, adaptive assessments, and real-time feedback, addressing diverse learning styles and preferences.

These benefits collectively contribute to a more streamlined educational environment that supports both administrative efficiency and personalized learning experiences for students.

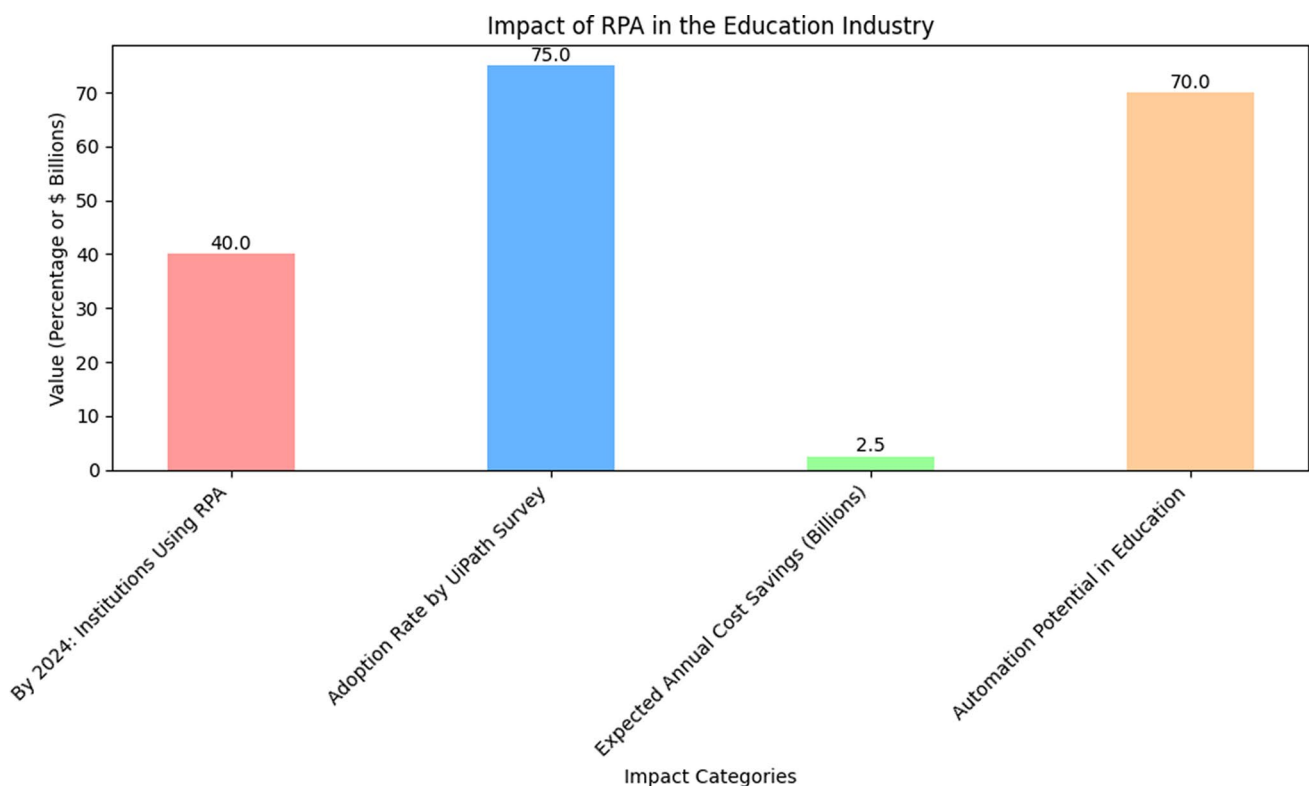
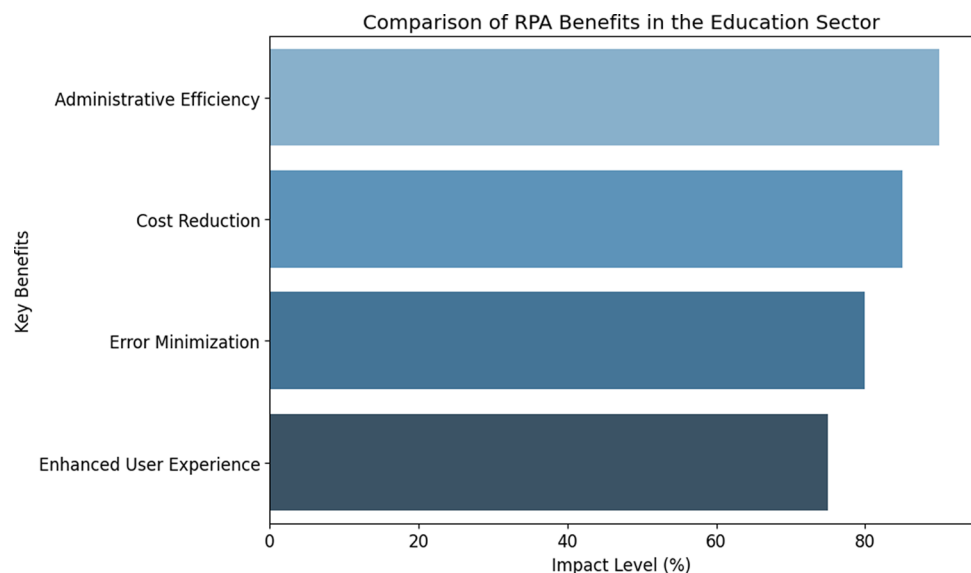


Fig. 8 Adoption Rates and Projected Cost Savings of RPA

Fig. 9 Key benefits of implementing RPA in educational settings



4.4 What challenges and ethical considerations have arisen from RPA adoption in education, and how are institutions addressing these issues?

While RPA provides substantial benefits to the education sector, several challenges and considerations must be addressed: **Implementation Costs:** The initial investment in RPA, including software and training expenses, can be considerable. However, long-term savings and increased efficiency generally justify these upfront costs [55]. **Data Security:** Since educational institutions handle sensitive student information, data security is paramount. Strong encryption and security measures are necessary to protect these data from potential breaches. **Change Management:** The introduction of RPA may necessitate changes in workflows and staff responsibilities. Effective change management and thorough training are essential to ensure a seamless transition for employees. **Scalability:** RPA solutions need to be scalable to adapt to changes in student enrollment, technological advancements, and institutional growth. Ensuring scalability helps educational institutions remain efficient and responsive to evolving needs.

RPA in education presents several challenges and ethical considerations that institutions must address to ensure successful integration. One of the primary ethical concerns is data privacy, particularly with the use of AI-driven systems in smart education. Issues related to bias in AI algorithms and the digital divide have raised questions about fairness and accessibility, as institutions transition from Industry 4.0 to 5.0 [56]. In addition, challenges related to faculty and student readiness have emerged, as both groups may lack the necessary skills to adapt to RPA technologies. Educational institutions must, therefore, foster a culture of continuous learning and adaptation to mitigate these readiness issues [56].

Implementation hurdles further complicate the transition to RPA in higher education, as staff training, and technological integration need to align with educational goals. This requires a coordinated effort to ensure that RPA systems not only enhance efficiency but also support broader institutional objectives [25]. Moreover, the integration of machine learning with RPA has been explored as a way to enhance teaching effectiveness, particularly during the pandemic, but this integration also brings new challenges in terms of ensuring ethical use and monitoring of students [28].

The use of robotic systems as intelligent tutors raises ethical questions regarding their effectiveness compared to human teachers and the potential implications for traditional teaching methods [24]. Additionally, the application of RPA in education must address social and emotional challenges, as teachers may struggle with the emotional disconnect caused by robotic interactions. Continuous learning and development programs for teachers are necessary to keep pace with these technological advancements [57].

Institutions must also consider the ethical dimensions of RPA in terms of employment and the potential displacement of staff. Ensuring data privacy and protecting sensitive student information are critical components

of RPA deployment, and institutions need to implement guidelines and frameworks to address these concerns proactively [39]. Moreover, transparency in how RPA systems manage student data is crucial, and institutions must establish regulations and oversight to ensure ethical use and mitigate risks associated with data handling [33, 38]. By addressing these challenges and ethical considerations through policy development, training programs, and clear regulatory frameworks, educational institutions can responsibly adopt RPA technologies to enhance both administrative and academic processes [40].

These considerations are critical for educational institutions looking to leverage RPA effectively while addressing potential challenges in implementation and operation.

5 Limitations and recommendations for future research on education through RPA

RPA holds significant potential for enhancing educational environments, yet it also presents several key limitations that must be addressed for successful integration. One major challenge lies in the adaptability and complexity of RPA systems. These systems often struggle to automate complex tasks and process unstructured data, which are prevalent in dynamic educational settings [17]. As education is inherently fluid and diverse, RPA systems face difficulties in adapting to new tasks and environments and may fail to retain previously learned knowledge when encountering new scenarios. Additionally, the high costs of acquiring, implementing, and maintaining RPA systems can serve as a significant barrier to adoption for many educational institutions, especially those with limited financial resources [58]. Furthermore, the integration of RPA introduces technical complexities and raises concerns about data privacy and security, as well as the ethical implications of ensuring equitable access and preventing bias [16, 18]. Additionally, there is a lack of comprehensive research on RPA's application in educational settings, particularly regarding its impact on the learning process rather than just administrative tasks. The gap in educators' knowledge and training further hampers the effective use of RPA in classrooms and instructional practices.

To address these limitations and unlock RPA's full potential, future research should focus on several key areas. First, empirical studies and case studies can provide valuable insights into the practical impact of RPA on educational outcomes and best practices for its integration. Furthermore, the integration of advanced technologies like Artificial Intelligence (AI), Machine Learning (ML), and Natural Language Processing (NLP) could enhance RPA's capabilities, enabling systems to adapt more effectively to diverse educational environments [14, 15]. Educators also require ongoing professional development to bridge the existing knowledge gap and enable them to leverage RPA in their teaching practices. Additionally, exploring cost-effective and scalable solutions for RPA implementation can help make this technology more accessible to a broader range of institutions. Finally, developing ethical and security frameworks to address data privacy concerns is critical to ensuring that RPA systems are deployed responsibly, with appropriate safeguards in place to protect sensitive educational data.

6 Conclusion

In conclusion, Robotic Process Automation (RPA) has emerged as a transformative force in higher education, significantly enhancing operational efficiency and driving innovation across both administrative and teaching functions. Case studies from institutions like the University of Melbourne and the University of Auckland highlight the tangible benefits of RPA, including substantial time savings, reduced errors, and higher satisfaction rates for both students and staff. By automating repetitive tasks, RPA has not only streamlined administrative processes but also enabled more effective resource management and personalized learning experiences. However, challenges related to implementation costs, data security concerns, and the need for robust change management strategies remain obstacles to widespread adoption. To fully realize RPA's potential, educational institutions must focus on scalability, invest in ongoing staff training, and ensure that their infrastructures are equipped to handle these advanced technologies. As RPA continues to evolve alongside innovations in natural language processing (NLP) and machine learning (ML), its applications in education will only become more sophisticated, further enhancing its role in creating student-centered learning environments.

Collaboration between educational institutions, policymakers, and technology providers will be essential to address the ethical implications of RPA adoption, including concerns around job displacement and data privacy. Institutions that embrace RPA today will be well-positioned to lead the way toward a more efficient, agile, and student-focused educational system in the digital era.

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Declarations

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